Stochastic Portfolio Analysis: Multi-Asset Monte Carlo Simulation on BIST-100 Stocks

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Abstract

This report presents a quantitative framework for modeling financial market uncertainty by implementing Geometric Brownian Motion (GBM) and Cholesky Decomposition. By conducting a 1,000,000-scenario Monte Carlo simulation on Borsa İstanbul assets (THYAO, TUPRS, BIMAS), the study quantifies the mathematical impact of portfolio diversification on risk mitigation. Key results indicate a median portfolio growth of 63.98% over a 1-year horizon, with only a 2.00% probability of capital loss on a 300,000 TL initial investment.

1 Introduction

In financial engineering, rather than predicting point-wise future values, modeling the probability distributions of asset returns is crucial for robust risk management. This study applies stochastic differential equations—commonly used in signal processing and random process theory—to model the price trajectories of financial instruments listed on Borsa İstanbul (BIST-100).

2 Theoretical Framework

2.1 Geometric Brownian Motion (GBM)

Asset prices (S_t) are assumed to follow a log-normal distribution, modeled by the following stochastic differential equation:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t \tag{1}$$

Where μ represents the expected annual return (drift), σ denotes the annualized volatility, and dW_t is the Wiener process increment. The discretized Euler-Maruyama solution used in simulation is:

$$S_{t+\Delta t} = S_t \cdot \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_t\right] \tag{2}$$

where $Z_t \sim \mathcal{N}(0,1)$ and $\Delta t = 1/252$.

2.2 Cholesky Decomposition and Correlation

The primary challenge in multi-asset modeling is reflecting the inter-asset correlation into random shocks. Given the historical correlation matrix Σ, Cholesky decomposition yields a lower triangular matrix L such that:

$$LL^T = \Sigma \tag{3}$$

Independent standard normal shocks $X \sim \mathcal{N}(0, I)$ are transformed into correlated shocks via $Z = L \cdot X$, ensuring the simulated paths respect historical market co-movements.

3 Simulation Setup and Data

3.1 Asset Parameters

Historical daily log-returns were computed over a 1-year window retrieved via `yfinance`. Annualized drift and volatility parameters are presented in Table 1.

Table 1: Annualized Asset Parameters (1-Year Historical Window)

Asset	**Sector**	**Initial Price (TL)**	**Annual Return μ**	**Volatility σ**
THYAO	Aviation	746.00	53.9%	33.5%
BIMAS	Retail	317.25	1.8%	31.7%
TUPRS	Energy	268.00	85.0%	35.1%

3.2 Correlation Structure

Table 2 presents the historical pairwise correlation matrix of daily log-returns. The notably low correlation between TUPRS and the other assets (0.08–0.16) is a key driver of diversification benefits.

Table 2: Pairwise Correlation Matrix of Daily Log-Returns

	BIMAS	**THYAO**	**TUPRS**
BIMAS	1.000	0.504	0.165
THYAO	0.504	1.000	0.080
TUPRS	0.165	0.080	1.000

3.3 Simulation Parameters

The simulation was implemented in Python (`numpy`, `pandas`, `yfinance`) with the following configuration: 1,000,000 Monte Carlo scenarios, a 252-trading-day horizon ($T = 1$ year), and equal capital allocation of 100,000 TL per asset from a total portfolio of 300,000 TL.

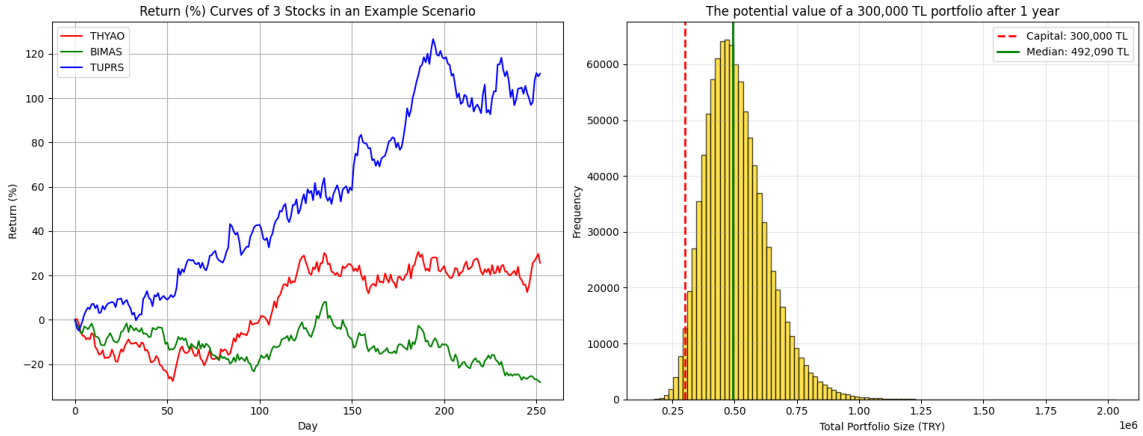

Figure 1: Left: Simulated return trajectories of THYAO, BIMAS, and TUPRS for a single representative scenario. Right: Distribution of total portfolio value after 1 year across 1,000,000 simulations. The red dashed line marks the initial capital (300,000 TL); the green line marks the median outcome (491,949 TL).

4 Results and Risk Analysis

Table 3 summarizes the key risk metrics derived from the 1,000,000-scenario simulation.

Table 3: Portfolio Risk Metrics (1,000,000 Scenarios, $T = 1$ Year)

Metric	**Value**
Initial Capital	300,000 TL
Median Portfolio Value	491,949 TL
Median Return	+63.98%
Probability of Capital Loss	2.00%
Probability of ≥50% Growth	64.34%
95% Value-at-Risk (VaR) Lower Bound	330,886 TL
99% Value-at-Risk (VaR) Lower Bound	280,992 TL

The 95% VaR result of 330,886 TL implies that in 95% of all simulated scenarios, the portfolio retains or exceeds its initial value — a direct consequence of the high expected drift components of THYAO ($\mu = 53.9\%$) and TUPRS ($\mu = 85.0\%$).

5 Conclusion

This study demonstrates a rigorous stochastic simulation pipeline for multi-asset portfolio analysis. The Monte Carlo results confirm that a diversified BIST-100 portfolio — constructed from low-correlated assets across aviation, energy, and retail sectors — exhibits substantially reduced tail risk. The 99% VaR bound of 280,992 TL indicates that even under extreme adverse scenarios, capital loss remains limited to approximately 6.3% of the initial investment.

From an engineering perspective, this framework is directly analogous to multi-channel noise modeling: Cholesky decomposition serves as the decorrelating transform, and port-

folio diversification functions as a matched filter that optimizes the Signal-to-Noise Ratio (SNR) of the investment strategy by exploiting low inter-channel (inter-asset) correlation.

6 References

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